

HIREWISE

Software Requirements Specification

AI-Powered Tech Recruitment & Interview Scheduling Platform

Document Version	1.0
Status	Baseline
Prepared For	HireWise Product & Engineering Team
Primary Client	React Web + Flutter Mobile
Deployment	AWS

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements for HireWise, an AI-powered platform for technical recruitment and interview scheduling. The document provides a shared baseline for product owners, developers, QA engineers, architects, security reviewers, and stakeholders.

1.2 Product Goal

HireWise shall streamline technical recruitment by bringing job management, candidate applications, AI-assisted evaluation, personalized interview preparation, interview scheduling, interviewer feedback, notifications, and recruitment analytics into one controlled platform.

1.3 SRS Conventions

Requirement identifiers use the following prefixes: FR for functional requirements, NFR for non-functional requirements, SEC for security requirements, DATA for data requirements, INT for integrations, AI for Agentic AI requirements, UI for interface requirements, and DEP for deployment requirements. Priority values are Must, Should, or Could.

1.4 Definitions of Priority

Priority	Meaning
Must	Required for the product to satisfy the defined baseline.
Should	Important and expected, but can be deferred only with an approved scope decision.
Could	Useful enhancement that does not block the baseline release.

2. Product Overview

2.1 Product Perspective

HireWise is a multi-client recruitment platform composed of a React web application, a Flutter mobile application, an ASP.NET Core business API, a PostgreSQL data layer hosted by Supabase, and a separate FastAPI + LangGraph AI service. Clerk provides identity and authentication services. The production runtime is hosted on AWS.

Architectural boundary: The client applications communicate with the ASP.NET Core business API. The ASP.NET Core service is the authoritative business layer and calls the internal AI service when an Agentic AI workflow is required. Clients must not directly invoke the AI service.

2.2 High-Level Architecture

Layer	Responsibility	Technology
Web Client	Recruiter, interviewer, administrator and optional candidate workflows	React + TypeScript + TailwindCSS + shadcn/ui
Mobile Client	Candidate-first mobile experience and mobile recruitment workflows	Flutter + Dart
Business API	Authentication enforcement, authorization, business rules, validation, persistence coordination and external-service orchestration	ASP.NET Core Web API
AI Service	Multi-agent planning, delegation, tool use, validation and workflow orchestration	FastAPI + LangGraph
Database	Business data, workflow state, audit data and reporting data	PostgreSQL on Supabase
Identity	Authentication and trusted identity lifecycle	Clerk
Production Infrastructure	Containerized runtime, secrets, networking, monitoring and CI/CD	AWS + Docker + Terraform + GitHub Actions

2.3 Core User Experiences

- Candidate: discover jobs, manage profile/resume, apply, track applications, view interviews and receive notifications.
- Recruiter: manage jobs and candidates, trigger and review AI evaluation, approve progression, manage interviews, view questions, feedback, analytics and audit summaries.
- Interviewer: view assigned interviews, candidate context, approved interview questions and submit structured feedback.
- Administrator: manage users and system-level settings, review audit logs and monitor platform-level activity.

3. Actors, Roles and Permissions

HireWise shall use four primary application roles. There is no separate Hiring Manager role; hiring-manager capabilities belong to Recruiter.

Role	Primary Responsibilities	Key Permissions
Candidate	Find jobs, maintain profile, apply and track progress	View published jobs; manage own profile/resume; create/withdraw permitted applications; view own application/interview status; access own notifications
Recruiter	Own the recruitment process and hiring-manager capabilities	Manage company/job data; review candidates; trigger AI workflows; approve/reject/revise recommendations; schedule interviews; manage questions; review feedback; access analytics
Interviewer	Conduct interviews and submit feedback	View assigned interviews; view relevant candidate/job information; access approved question sets; submit/edit feedback for assigned interviews
Admin	Platform administration and oversight	Manage users/roles; view audit records; manage platform settings; access operational analytics

3.1 Permission Principles

- Least privilege: users receive only permissions required for their role.
- Server-side authorization is authoritative; client-side visibility is not a security control.
- Candidate data is scoped to the candidate or authorized recruitment users.
- Interviewers can only access interviews and candidate information required for their assigned responsibilities.
- Sensitive administrative operations require explicit authorization and audit logging.

4. System Scope and Boundaries

4.1 In Scope

- Organization, department and technical job management.
- Candidate profiles, resumes, education, experience, projects and certifications.
- Job discovery, applications and application lifecycle tracking.
- AI-assisted job analysis, resume analysis, candidate evaluation/ranking, interview question generation and interview scheduling.
- Recruiter review and human approval of high-impact AI recommendations.
- Interviewer availability, interviews, feedback and interview history.
- Notifications, dashboards, reporting, analytics and audit summaries.
- React web and Flutter mobile clients using the same identity, data and business rules.
- Authentication, authorization, structured logging, validation, security controls and testing.
- AWS production deployment with Docker, Terraform and GitHub Actions.

4.2 Out of Scope

- Autonomous final hiring decisions without recruiter approval.
- Payroll, employee onboarding, attendance or general HR management after hiring.
- A full video-conferencing product implemented inside HireWise.
- Training a foundation language model.
- A native Flutter-only business backend separate from the shared platform.

- Client applications calling the AI service directly.

5. Functional Requirements

ID	Requirement	Description	Priority
FR-001	Authentication	The system shall allow users to sign in and sign out using Clerk-managed authentication.	Must
FR-002	Identity Synchronization	The system shall synchronize trusted user identity and role information from Clerk into the application user model.	Must
FR-003	Role Authorization	The system shall enforce role-based access for Candidate, Recruiter, Interviewer and Admin operations.	Must
FR-004	Company Management	Authorized Recruiters and Admins shall create, view, update and manage organization and department records.	Must
FR-005	Job Creation	Recruiters shall create technical job postings with role title, description, skills, experience, education and status information.	Must
FR-006	Job Publishing	Recruiters shall publish, pause and close jobs subject to business validation.	Must
FR-007	Job Discovery	Candidates shall search and filter published jobs.	Must
FR-008	Candidate Profile	Candidates shall create and maintain their profile, education, experience, project and certification information.	Must
FR-009	Resume Management	Candidates shall upload and manage their resumes subject to file validation and access controls.	Must
FR-010	Applications	Candidates shall submit applications to eligible open jobs and view their application history.	Must
FR-011	Application Validation	The system shall reject invalid or duplicate applications and provide a meaningful error message.	Must
FR-012	Application Lifecycle	The system shall maintain controlled application status transitions and history.	Must
FR-013	Candidate Review	Recruiters shall search, filter and review candidates and applications.	Must
FR-014	AI Evaluation Initiation	Recruiters shall be able to start a defined AI recruitment workflow for an eligible application.	Must
FR-015	AI Recommendation Review	Recruiters shall view structured AI recommendations and approve, reject or request revision.	Must
FR-016	Interview Approval	Only an authorized Recruiter shall be able to approve progression to the interview stage.	Must
FR-017	Interview Question Generation	The system shall generate a personalized interview question set using job and candidate context after interview approval.	Must
FR-018	Interviewer Availability	Interviewers shall manage available scheduling windows subject to application rules.	Must
FR-019	Interview Scheduling	Recruiters shall schedule, reschedule or cancel interviews while preventing conflicts.	Must
FR-020	Interview Details	Candidates and assigned Interviewers shall view interview details appropriate to their role.	Must
FR-021	Interview Feedback	Assigned Interviewers shall submit structured feedback and update the interview outcome.	Must
FR-022	Notifications	The system shall provide relevant recruitment and interview notifications to users.	Must
FR-023	Recruitment Analytics	Recruiters and Admins shall access recruitment pipeline and operational analytics.	Must
FR-024	AI Monitoring	Authorized users shall view AI workflow status, execution summaries, validation results and approval decisions.	Must
FR-025	Audit History	The system shall record important recruitment, security, AI and approval actions for traceability.	Must
FR-026	Cross-Platform Synchronization	React and Flutter clients shall use the same identity, permissions, data and business rules.	Must

6. Detailed Business Workflows

6.1 Candidate Registration and Onboarding

1. Candidate selects sign-in/registration.
2. Clerk performs identity verification.
3. HireWise receives the authenticated identity and resolves the application role.
4. Candidate completes required profile information.
5. Candidate uploads a resume if applying to jobs requiring one.
6. The system validates profile and document data.

6.2 Job Publishing

7. Recruiter creates a draft job.
8. Recruiter adds required and preferred skills plus relevant job criteria.
9. System validates required fields and business constraints.
10. Recruiter publishes the job.
11. Published job becomes available to candidates for discovery.

6.3 Candidate Application

12. Candidate views a published job.
13. Candidate selects Apply.
14. System checks job status and duplicate-application rules.
15. Candidate submits the application and required information.
16. System stores the application and creates an application history record.
17. Recruiter receives or views the new application.

6.4 AI Candidate Evaluation

18. Recruiter starts the AI workflow for an eligible application.
19. The workflow loads the job requirements and candidate information.
20. Job Analysis Agent structures job criteria.
21. Resume Analysis Agent structures candidate information.
22. Candidate Evaluation Agent compares job and candidate information.
23. Validation Agent performs deterministic checks.
24. HireWise stores the recommendation and execution summary.
25. The workflow pauses for Recruiter approval when a high-impact action is reached.
26. Recruiter may approve, reject or request revision.

6.5 Interview Preparation and Scheduling

27. Recruiter approves the candidate for interview.
28. Interview Question Generator Agent creates a personalized question set.
29. The system stores the approved question set with its context.
30. Interviewer availability and candidate availability are considered.
31. Scheduling Agent proposes suitable slots.
32. Deterministic conflict validation runs.
33. Recruiter confirms the schedule.
34. The system creates the interview and sends notifications.

6.6 Interview Completion

35. Interviewer views the assigned interview.
36. Interviewer reviews candidate context and approved questions.
37. Interviewer records structured feedback.
38. System stores feedback and updates interview status.
39. Recruiter reviews feedback and continues the candidate lifecycle.

7. Agentic AI Requirements

HireWise shall use a controlled multi-agent workflow rather than a single prompt or generic chatbot. The AI subsystem shall be isolated as an internal service and shall expose structured workflow operations to the ASP.NET Core business API.

7.1 Agent Inventory

Agent	Responsibility	Inputs	Outputs
Job Description Analysis Agent	Extract and normalize job requirements	Job description, structured job fields	Required skills, preferred skills, experience and qualification criteria
Resume Analysis Agent	Extract candidate evidence from resumes and profile data	Resume/document content, profile data	Structured skills, education, experience, projects and certifications
Candidate Evaluation & Ranking Agent	Compare candidate evidence with job requirements	Job criteria, candidate profile	Scores, strengths, gaps, recommendation and rationale summary
Validation Agent	Validate outputs and business-rule compliance	Structured agent outputs, application/job state	Validation result, violations, revision/safe-failure decision
Interview Question Generator Agent	Create candidate-specific interview questions	Job requirements, candidate profile, interview type	Structured technical, project, problem-solving and behavioral questions
Interview Scheduling Agent	Recommend feasible interview slots	Candidate/interviewer availability, duration, scheduling rules	Candidate slots, conflict checks and recommended schedule

7.2 AI Workflow State

- Each workflow shall have a unique workflow identifier.
- Workflow state shall include objective, plan, current step, completed steps, structured outputs, tool results, validation outcomes, approval state, errors and final outcome.
- Workflow state shall be durable and recoverable.
- Hidden model reasoning shall not be persisted.

7.3 AI Tool Control

- Each agent shall receive only the tools necessary for its responsibility.
- Tool inputs shall be validated before execution.
- Tool outputs shall be validated against expected schemas.
- Tool execution shall be recorded in an auditable summary.
- Failed tools shall return controlled errors and must not cause uncontrolled cascading actions.

7.4 Human Approval

High-impact recruitment actions shall pause the workflow until an authorized Recruiter explicitly approves, rejects or requests revision. Approval decisions shall be recorded with actor, timestamp and optional comment.

7.5 Safe Failure

If the AI workflow encounters invalid output, unavailable tools, policy violations, unsupported inputs, repeated errors or timeout limits, it shall stop safely, preserve the relevant workflow state and expose a recoverable failure status to authorized users.

8. User Interface Requirements

8.1 React Web Application

Area	Primary Users	Required Capabilities
Authentication	All roles	Sign-in, sign-out, protected navigation and session state
Recruiter Workspace	Recruiter	Jobs, applications, candidate review, AI evaluation, approvals, interviews, feedback, analytics
Interviewer Workspace	Interviewer	Assigned interviews, candidate context, question sets, feedback

Area	Primary Users	Required Capabilities
Admin Workspace	Admin	User management, audit records, operational settings and analytics
Candidate Web Experience	Candidate	Job discovery, profile, applications and status tracking where enabled

8.2 Flutter Mobile Application

Flutter shall provide a genuine mobile experience over the same HireWise business API. The mobile application shall prioritize candidate-facing and operational workflows rather than duplicating every administrative screen from the web application.

- Registration and login.
- Job discovery and search.
- Job details and application submission.
- Profile and resume management.
- Application status and history.
- Interview details and updates.
- Notifications.
- Secure token/session handling.
- Responsive mobile layouts and clear loading/error states.

8.3 Shared Client Behavior

- Both clients shall display status values consistently.
- Both clients shall respect the same role and permission rules.
- Business decisions shall not be duplicated as independent client-only rules.
- Validation messages should be clear and actionable.
- Loading, empty, success and error states shall be represented for network-dependent operations.

9. External Interface Requirements

9.1 API Interface

The ASP.NET Core API shall expose RESTful endpoints for all public business operations. Endpoints shall use standard HTTP methods and status codes, request/response DTOs, validation and consistent error responses. Swagger/OpenAPI documentation shall be generated for the business API.

Interface Area	Examples	Interface Requirement
Authentication	Clerk JWT	API validates token and user identity before protected operations.
Jobs	/api/jobs	Create, publish, update, search and retrieve jobs according to authorization.
Applications	/api/applications	Submit, retrieve, filter and update application status according to rules.
AI Workflows	/api/ai/workflows	Start workflow, view status, approve, reject, revise and retrieve execution summary.
Interviews	/api/interviews	Manage slots, schedule, reschedule, cancel and retrieve interviews.
Audit	/api/admin/audit-logs	Authorized users retrieve audit summaries.

9.2 AI Service Interface

FastAPI endpoints shall be internal-facing and called by ASP.NET Core only. The AI service shall not become a public browser-facing API.

Operation	Purpose
Start Workflow	Begin a controlled recruitment workflow.
Get Workflow	Return current structured workflow state and execution summary.
Resume/Revise Workflow	Continue a workflow after a valid revision or approval action.

Operation	Purpose
Cancel Workflow	Safely terminate an in-progress workflow when authorized.

9.3 Third-Party Integrations

Integration	Purpose	Requirements
Clerk	Identity and authentication	Use secure tokens; use trusted server-side webhook handling for role synchronization; do not expose secrets.
Calendar Provider	Interview scheduling	Create/update/cancel events through controlled backend integration; handle conflicts, failures and rate limits.
Email/Notification Provider	Recruitment and interview notifications	Send template-driven messages; handle transient failure and avoid leaking sensitive candidate information.
Cloud Storage	Resume/document storage if adopted	Private storage, access-controlled retrieval, file validation and lifecycle management.

10. Data Requirements

10.1 Core Data Domains

Domain	Key Data Objects
Identity & Access	User, Role, Permission, UserRole
Organization	Company, Department
Jobs	Job, Skill, JobSkill
Candidates	CandidateProfile, Resume, Education, Experience, Project, Certification
Recruitment	Application, ApplicationStatusHistory, CandidateEvaluation, CandidateScore, Recommendation
Interviews	Interview, InterviewSlot, InterviewerAvailability, InterviewParticipant, InterviewFeedback, InterviewQuestionSet, InterviewQuestion
AI Workflow	AIWorkflow, AIWorkflowStep, AgentExecution, ToolExecution, ValidationResult, ApprovalRecord
Platform	Notification, AuditLog

10.2 Data Quality Requirements

- Primary keys shall be unique.
- Foreign key relationships shall preserve referential integrity.
- Required business fields shall not accept invalid null or empty values.
- Duplicate application and conflicting schedule rules shall be enforced.
- Important timestamps shall be stored consistently.
- Business status transitions shall be controlled rather than stored as arbitrary strings.
- The database shall support indexes for common search, filtering and status queries.

10.3 Data Retention and Privacy

Candidate documents and personal information shall be stored only for legitimate platform purposes. Retention and deletion rules shall be configurable where required. Access to candidate data shall follow role permissions and minimum-necessary access.

11. Security Requirements

ID	Area	Requirement
SEC-001	Authentication	All protected operations shall require a valid Clerk-authenticated identity.
SEC-002	Authorization	The server shall enforce role-based and resource-level permissions.

ID	Area	Requirement
SEC-003	Role Integrity	Users shall not be able to self-assign privileged roles from the client application.
SEC-004	Webhook Security	Clerk webhook requests shall be verified before role synchronization is performed.
SEC-005	Transport Security	Production client/API and service-to-service traffic shall use encrypted transport.
SEC-006	Secrets	Secrets shall be stored outside source control and injected securely through the deployment environment.
SEC-007	Input Validation	API, file, tool and AI inputs shall be validated before processing.
SEC-008	File Security	Resume uploads shall enforce file type, size and content validation and use private storage.
SEC-009	AI Tool Security	Agents shall use allow-listed tools with least-privilege access.
SEC-010	Prompt Injection Resistance	External text such as resumes and job descriptions shall be treated as untrusted content and shall not be allowed to redefine tool permissions or system rules.
SEC-011	Audit	Security-sensitive and high-impact business actions shall create audit records.
SEC-012	Error Disclosure	Error responses shall not expose stack traces, secrets or internal infrastructure details to clients.

12. Non-Functional Requirements

ID	Area	Requirement	Priority
NFR-001	Performance	Common API operations should provide responsive interactions under normal expected load. AI workflows shall expose visible progress/state rather than blocking indefinitely.	Should
NFR-002	Availability	Production services should use appropriate AWS managed services and health checks to support reliable availability.	Should
NFR-003	Scalability	The architecture shall allow web, API and AI workloads to scale independently where practical.	Should
NFR-004	Reliability	Important operations shall be idempotent where practical and preserve recoverable state on failure.	Must
NFR-005	Usability	Primary workflows shall have clear navigation, feedback, loading and error states.	Must
NFR-006	Accessibility	Web interfaces shall follow practical accessible design patterns, keyboard navigation and readable contrast.	Should
NFR-007	Maintainability	Business logic, AI agents, tools and validation shall remain modular and independently testable.	Must
NFR-008	Observability	Applications shall produce structured logs and meaningful health/status information.	Must
NFR-009	Auditability	High-impact business and AI actions shall be traceable.	Must
NFR-010	Compatibility	The web application shall support current mainstream desktop browsers; Flutter shall support the targeted mobile platform.	Should
NFR-011	Recoverability	AI and integration failures shall result in safe states that can be retried, revised or manually resolved.	Must
NFR-012	Deployability	Services shall be containerized and deployable through the defined AWS/Terraform/GitHub Actions pipeline.	Must

13. Logging, Monitoring and Auditability

13.1 Application Logging

Logs shall be structured and include, where relevant, timestamp, service, environment, correlation ID, request ID, authenticated user ID, role, operation, outcome and execution duration. Sensitive secrets and unnecessary personal data shall not be logged.

13.2 AI Execution Logging

Recorded Item	Requirement
Workflow ID	Every AI workflow execution shall be traceable to a unique workflow.

Recorded Item	Requirement
Agent	The agent responsible for a step shall be identified.
Tool	Tool calls shall be summarized with timing and outcome.
Validation	Validation results and rule failures shall be recorded.
Approval	Approval actor, decision and timestamp shall be recorded.
Failure	Errors, retries and safe-failure outcomes shall be recorded.

13.3 Audit Events

- Authentication and role changes.
- Job publication/closure.
- Application status changes.
- AI workflow initiation and completion.
- Recruiter approvals and rejections.
- Interview creation/reschedule/cancel.
- Interview feedback submission.
- Administrative configuration changes.

14. Validation and Error Handling

14.1 Validation Layers

Layer	Responsibility
Client	Immediate form and presentation validation; not the authoritative security/business layer.
ASP.NET API	Request validation, authorization, business rules and resource checks.
Domain	Core business invariants and state-transition rules.
Database	Constraints, relationships and data integrity.
AI Service	Structured output validation, tool-input validation and workflow-specific AI checks.

14.2 Error Response Behavior

- Errors shall use consistent, machine-readable codes.
- User-facing messages shall explain what can be corrected without exposing internal details.
- Validation failures shall not partially commit invalid state.
- External-service failures shall be retryable when safe.
- AI failures shall preserve workflow state and expose safe-failure status.

15. Testing and Quality Requirements

Test Area	Required Coverage
Backend Unit Tests	Domain rules, application services, validators and authorization policies
API Integration Tests	HTTP routes, status codes, authentication, authorization and persistence
Database Tests	Migrations, constraints, relationships, transactions and query behavior
React Tests	Components, forms, protected routes, API states and critical workflows
Flutter Tests	Widgets, forms, navigation, API integration, state and critical mobile workflows
AI Tests	Golden cases, structured-output assertions, deterministic validation, prompt-injection cases, approval enforcement, failure recovery
End-to-End Tests	Candidate application -> recruitment evaluation -> approval -> interview preparation -> scheduling
Performance Tests	API concurrency, response times, database response and AI workflow latency

Test Area	Required Coverage
Security Tests	Authorization bypass, role tampering, file upload validation, input handling and AI tool permission checks

15.1 Quality Gates

- Pull requests shall pass automated tests before merge.
- Build failures shall block deployment.
- Critical security findings shall be resolved or explicitly accepted before production release.
- Core workflow regressions shall fail CI.
- AI workflow changes shall include corresponding evaluation cases.

16. Deployment and Infrastructure Requirements

16.1 Production Platform

Production infrastructure shall use AWS. The deployment shall remain containerized and infrastructure-as-code managed through Terraform. The concrete AWS services may be selected during the technical design phase, but the deployment shall support independent operation of the web application, business API and AI service.

Concern	AWS-Oriented Requirement
Container Runtime	Run web, ASP.NET Core and FastAPI workloads on an appropriate AWS container platform such as ECS/Fargate or an equivalent approved managed container service.
Container Registry	Store versioned Docker images in Amazon ECR.
Secrets	Use AWS Secrets Manager or Parameter Store for runtime secrets.
Monitoring	Use Amazon CloudWatch for logs, metrics, alarms and service visibility.
Networking	Use secure VPC/network controls and private communication for internal services where appropriate.
Ingress	Expose only required public endpoints; keep the AI service internal.
CI/CD	GitHub Actions shall build, test, scan, publish and deploy versioned artifacts.
Infrastructure as Code	Terraform shall manage the reproducible AWS infrastructure configuration.

16.2 Supabase

PostgreSQL shall continue to be hosted on Supabase as the managed application database. The production deployment shall use secure database credentials, restricted access and migration-controlled schema changes.

16.3 Environment Separation

- Local development environment.
- Shared/staging environment.
- Production environment.
- Environment-specific configuration shall be injected through secure environment variables or secret management rather than committed to source control.

17. Traceability and Acceptance

17.1 BRD-to-SRS Traceability

Business Area	Representative SRS Coverage
Job & Company Management	FR-004 to FR-007, data requirements, permissions and UI requirements
Candidate & Application Management	FR-008 to FR-013, data, file security, mobile and web requirements
Recruitment Intelligence	FR-014 to FR-016, AI sections, approval and audit requirements
Interview & Scheduling	FR-017 to FR-022, integration and conflict-validation requirements

Business Area	Representative SRS Coverage
Cross-Platform Experience	FR-026, UI requirements and shared permission/data behavior
Security & Quality	SEC-001 to SEC-012, NFRs, testing and observability requirements
Deployment & Operations	DEP-001 style expectations represented in Section 16, CI/CD and monitoring requirements

17.2 Baseline Acceptance Criteria

ID	Area	Acceptance Criterion
AC-01	Authentication & Roles	A user can authenticate, and the system exposes only role-appropriate operations.
AC-02	Job Lifecycle	A Recruiter can create, validate, publish and close a job.
AC-03	Candidate Application	A Candidate can discover a job, apply once, and track the application.
AC-04	AI Recruitment Workflow	The system can execute the multi-agent recruitment workflow with structured state, tools, validation and safe failure.
AC-05	Human Approval	A high-impact recruitment action pauses until Recruiter approval.
AC-06	Interview Questions	An approved candidate receives a personalized question set associated with the job and interview context.
AC-07	Interview Scheduling	The system prevents scheduling conflicts and records a confirmed interview.
AC-08	Interview Feedback	An assigned Interviewer can submit structured feedback.
AC-09	Cross-Platform	The same recruitment state is visible consistently from the appropriate React and Flutter experiences.
AC-10	Auditability	Authorized users can review relevant AI execution and business audit summaries.
AC-11	Security	Unauthorized requests are rejected and role tampering is prevented.
AC-12	Deployment	The production stack can be deployed through the defined Docker/Terraform/AWS/GitHub Actions process.

18. Risks, Assumptions and Constraints

18.1 Risks

Risk	Impact	Mitigation
AI recommendation error	Poor candidate recommendations	Structured outputs, deterministic validation, golden tests and Recruiter approval
AI bias	Unfair candidate treatment	Use job-relevant criteria, monitor outputs and retain human oversight
Scheduling conflict	Missed or duplicated interviews	Availability checks and server-side conflict validation
Data privacy exposure	Candidate trust and compliance concerns	Least privilege, encryption, private storage and audit logs
External service outage	Scheduling or notification delays	Timeouts, retries, safe-failure states and manual recovery
Scope growth	Delayed or incomplete delivery	Prioritize core recruitment workflow and defer non-essential features
AI service latency/cost	Slow or expensive workflows	Model limits, workflow checkpoints, caching where safe and execution monitoring

18.2 Assumptions

- Clerk remains the identity provider.
- Supabase remains the PostgreSQL provider.
- AWS is the production cloud environment.
- Suitable email/calendar providers are available for integration.
- Users provide accurate job and candidate information.
- AI outputs are advisory and require validation and human control.

18.3 Constraints

- The platform must protect candidate information.
- The AI service must remain an internal service behind ASP.NET Core.
- Critical business rules must not depend solely on LLM behavior.
- The project must remain modular enough for independent testing and future maintenance.

19. Future Enhancements

- Advanced candidate sourcing and talent-pool management.
- Integrated coding assessments and automated technical test evaluation.
- Interview transcription and structured post-interview summaries.
- Advanced recruiter recommendations using historical hiring outcomes.
- Candidate communication automation with configurable templates.
- Additional calendar and collaboration integrations.
- Employer branding and public career-page management.
- More advanced recruitment analytics and forecasting.

20. Glossary

Term	Definition
Agentic AI	An AI system capable of planning, delegating tasks, using controlled tools, maintaining workflow state, validating outputs and completing multi-step objectives.
Candidate	A person applying for a technical job through HireWise.
Recruiter	The organization-side user responsible for the hiring workflow, including capabilities normally associated with a Hiring Manager.
Interviewer	A person assigned to evaluate a candidate through an interview.
AI Workflow	A controlled sequence of AI and business actions used to support a recruitment objective.
Human Approval	An explicit action by an authorized Recruiter before a defined high-impact step is finalized.
Safe Failure	A controlled workflow termination or pause that preserves state and avoids unsafe or invalid business actions.
Application	A Candidate submission to a specific published job.
Interview Question Set	A structured collection of questions generated or selected for a specific job/candidate interview.
Workflow State	Persisted structured information describing the progress, outputs, validation and approval state of an AI workflow.

Appendix A. Requirement ID Summary

Prefix	Coverage
FR	Functional requirements
AI	Agentic AI behavior and orchestration requirements
UI	User interface requirements
INT	External and service interfaces
DATA	Data and integrity requirements
SEC	Security requirements
NFR	Non-functional requirements
DEP	Deployment/infrastructure expectations
AC	Acceptance criteria

Change control: Material changes to roles, scope, core workflows, AI responsibilities, data ownership, authentication, or deployment architecture should be reflected in this SRS and related design documents before implementation changes are merged.